

DSC 335, Handout 07: Gamma-Poisson Conjugate Family (*Chapter 5.1-5.2*)

LEARNING OBJECTIVES.

- *Determine the posterior probability distribution in the case of conjugate priors.*

Recall: We say that $f(\pi)$ is a conjugate prior for the likelihood $L(\pi|y)$ if the posterior $f(\pi|y) \propto f(\pi)L(\pi|y)$ is from the same model family as the prior.

What happens if we try to use a non-conjugate prior?

Example 1. Suppose we have a parameter π which represents a proportion between 0 and 1. Consider the following prior distribution for π : $f(\pi) = e - e^\pi$ for $\pi \in [0, 1]$.

- Show that $f(\pi)$ is a legitimate pdf.
- Suppose we observe 10 out of 50 successes in independent trials. Write down the likelihood.
- Using the prior and likelihood, give the posterior distribution.

Poisson Distribution

In R: For a Poisson random variable X , the function `dpois(x, lambda)` gives $P(X = x)$ and the function `ppois(x, lambda)` gives $P(X \leq x)$.

Example 2. Suppose Dr. Daryl had a single office hour each day of the week, and that the number of students that attend office hours each day can be modeled by a $Pois(3)$ distribution. What is the probability that exactly 5 students show up to office hours tomorrow? less than 3 students?

Suppose that Dr. Daryl wants to model the number of students that show up to office hours each day in a 5 day week. Then instead of a single Y , he has Y_1, Y_2, Y_3, Y_4, Y_5 . If he assumes the number of students that attend office hours each day are independent, then for each day i he has

Joint probability mass function

Example 3. Write down and simplify the joint pmf of the number of daily students to Dr. Daryl's office hours in a week. Then use the joint pmf to define a likelihood function.

Gamma Distribution

In R: The `plot_gamma(s,r,mean=FALSE,mode=FALSE)` function in the `bayesrules` package in R plots the Gamma pdf for given parameter values.

Example 4. Plot the Gamma pdf with various parameter values. How does the shape, center, and spread of the distribution change with the parameters?

Example 5. Suppose that based on the last few semesters, Dr. Daryl thinks that an average rate of around 3 students per office hour is reasonable, and that it probably isn't higher than 4.5 hours or lower than 2 hours. Give a reasonable Gamma distribution that models this prior information.

Gamma-Poisson Bayesian Model

Deriving the Gamma-Poisson Model

In R: The `plot_gamma_poisson(shape, rate, sum_y, n)` in the `bayesrules` package plots by default the prior, scaled likelihood, and posterior distributions for the Gamma-Poisson model. The `summarize_gamma_poisson(shape, rate, sum_y, n)` in the `bayesrules` package gives summaries of the prior and posterior distributions.

Example 6. (a) Suppose that Dr. Daryl observes 0, 4, 3, 1, and 4 students in his office hours one week. Using the prior developed above, calculate the posterior distribution of the hourly rate of students attending office hours.

(b) Could you have arrived at the same answer without knowing the individual daily counts but just the weekly count?

(c) Suppose he continues counting 12 total each week for the first 4 weeks of the semester. How will that change the posterior distribution? Make a guess based on intuition first, then check your guess analytically.

Example 7. Environmental scientists are studying a certain type of disease that affects trees. Suppose the number of diseased trees in a one-acre section of a forest can be modeled with a Poisson distribution with unknown rate λ . Other forests nearby have been studied, and the scientist calculates the average rate to be around 30 diseased trees per acre, with a standard deviation of around 10. The scientist thinks a Gamma distribution is a reasonable way to model the unknown rate parameter. Determine a reasonable prior for λ . Then, supposing there are 242 diseased trees in a 10-acre sample of the new forest, calculate the posterior distribution of λ .